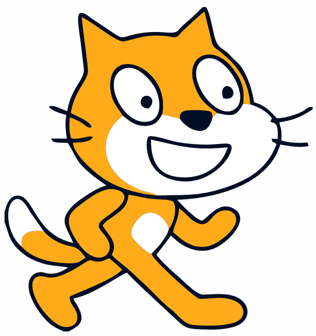
UNIT 2

INTRODUCTION TO SCRATCH

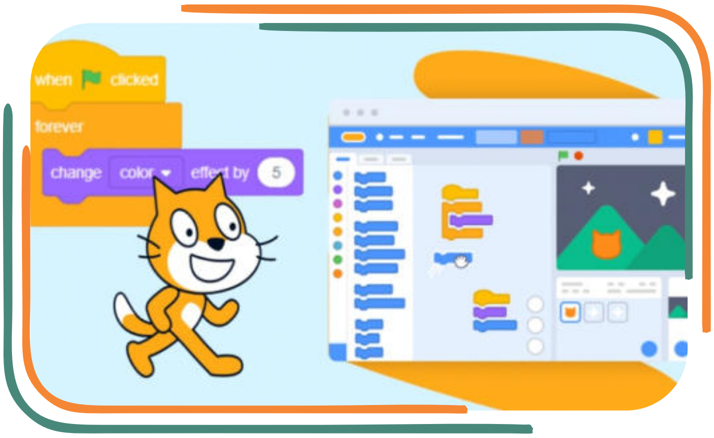


Learning Objectives

- Understand what Scratch is and why it's used.
- Learn the basic parts of the Scratch interface.
- Discover how to create and control characters (sprites).
- Start making simple animations and games.

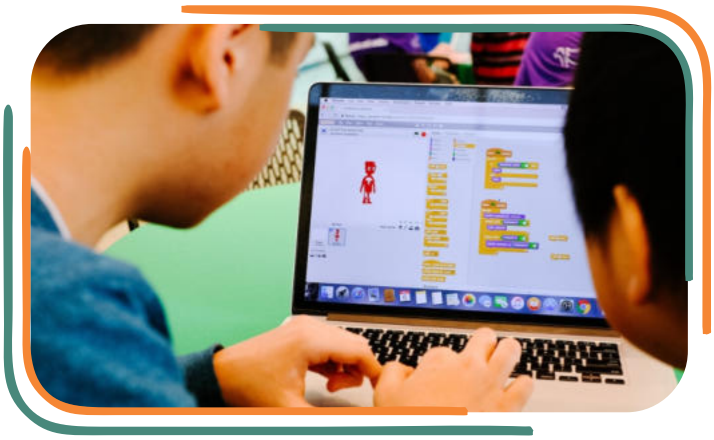
What is Scratch?

Scratch is a visual programming language designed to help beginners create interactive stories, games, and animations. With Scratch, you don't need to write long lines of code. Instead, you use colourful blocks that fit together like puzzle pieces, each containing predefined instructions for controlling characters called **sprites**. Scratch makes learning programming simple and fun, making it especially ideal for kids.



Why Use Scratch?

Learning Scratch helps you understand the basics of coding and problem-solving. Coding in Scratch also teaches you to think creatively, work on exciting projects, and see immediate results. As you progress, you'll find that coding can be both a useful skill and a lot of fun!



Getting Started with Scratch

1. Open a web browser (like Chrome or Firefox).
2. Go to the website: https://scratch.mit.edu.
3. Click on "Create" at the top of the page to start a new project.

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Understanding the Scratch Interface

The Scratch screen has several parts. Here's a quick guide:

Stage: This is the main area where you see your animations and characters in action. It's like a theater stage where all the scenes play out.

Sprite Pane: This shows all the sprites in your project. Each sprite is like a character or an object in your project, and you can add or remove sprites here.

Block Palette: This area has all the different blocks that you can use to control your sprites. Blocks are grouped by colour and function (e.g., Motion, Looks, Sound).

Script Area: This is the workspace where you build scripts by snapping blocks together. Scripts are sequences of instructions that tell the sprites what to do.

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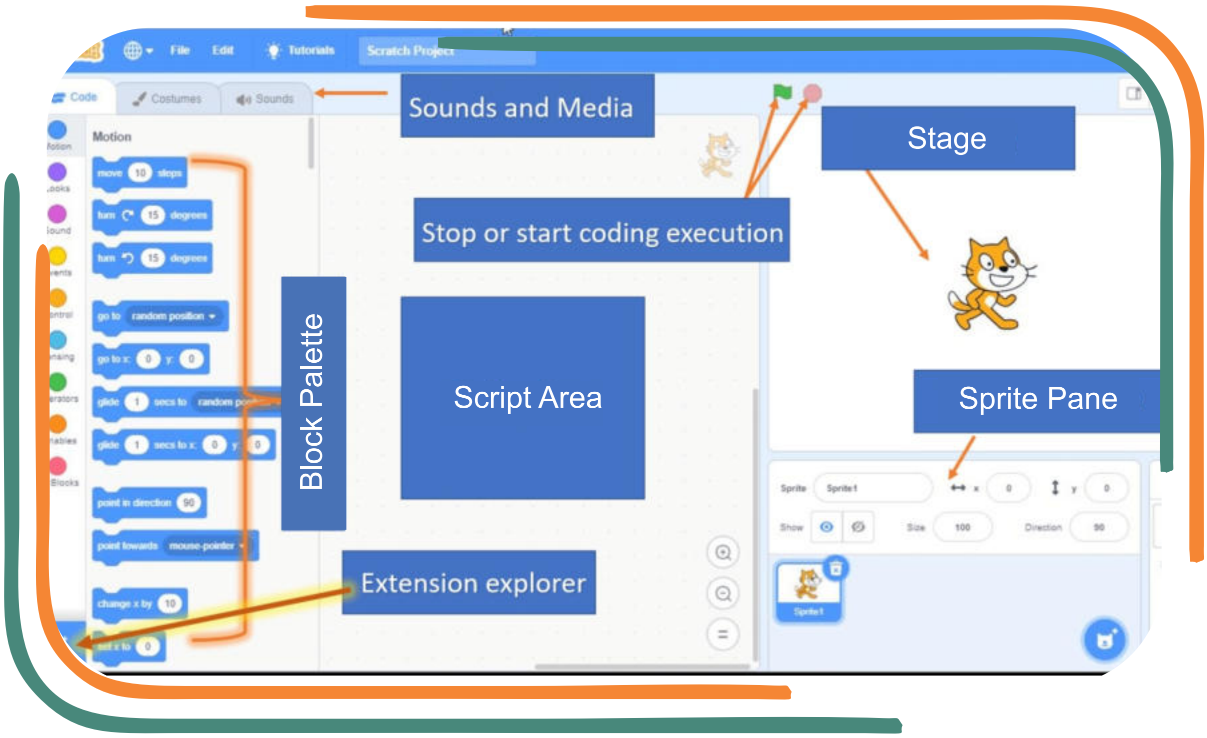

Key Terms in Scratch

Let's learn a few important terms:

Sprites: Sprites are characters or objects that you can move, change, and animate. Scratch has many sprites to choose from, or you can create your own.

Blocks: Blocks are the building blocks of Scratch coding. Each block has a specific function, like moving a sprite or playing a sound. You snap blocks together to create scripts.

Scripts: A script is a set of blocks that work together to perform an action. For example, you can create a script to make a sprite move across the screen.

Backdrop: A backdrop is the background of the stage. It sets the scene for your animation or game, and you can change it to create different settings.

Creating Your First Project

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Goal: Make a sprite move and say something.

1. Add a Sprite:

In the Sprite Pane, click on the choose a Sprite button. Select a sprite (like a cat) from the library.

2. Create a Script:

In the Block Palette, go to the **Events** category and drag the "**when green flag clicked**" block to the Script Area. This block starts your script when you click the green flag.

3. Add Movement:

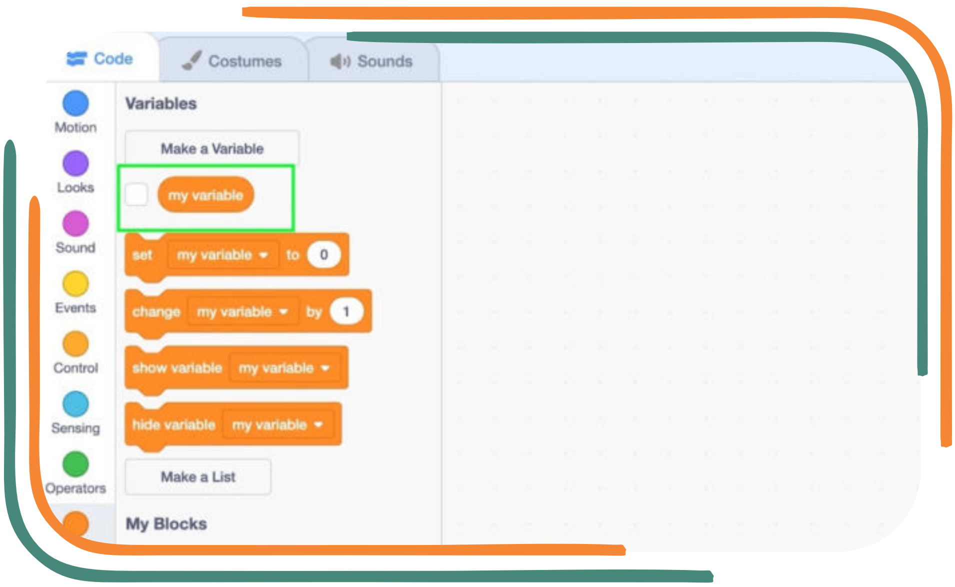
Go to the **Motion** category. Drag the "move 10 steps" block under the "when green flag clicked" block. Now, when you click the green flag, your sprite will move 10 steps forward.

4. Add Dialogue:

Go to the **Looks** category. Drag the "**say Hello**! for 2 seconds" block under the "move 10 steps" block. Change "Hello!" to whatever you want your sprite to say.

5. Test Your Script:

Click the green flag at the top of the Stage. Your sprite should move and say something!



Adding Backgrounds and Sounds

Changing the Backdrop:

1. In the Stage area, click on the Choose a Backdrop button.
2. Select a background you like, such as a beach or a city. This will change the backdrop on the stage.

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How to Draw Your Own Backdrop in Scratch

1. Open Scratch: Start a new project.

2. Go to the Backdrop Editor:

- Click on the Stage area (where the backdrop is).
- Click the "Backdrops" tab.
- Click "Paint" to open the drawing tools.

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3. Draw Your Backdrop:

- Use the tools on the left.
- Brush to draw freehand.
- Fill to colour areas.
- Rectangle and Circle for shapes.
- Line to draw straight lines.
- Choose your colours from the colour palette.

4. Save Your Backdrop: When you're done, click "OK" to save it.

5. Set the Backdrop:

- Use the "when green flag clicked" block to start your backdrop in the project.
- Go to Looks and drag "switch backdrop to [your backdrop name]". Now you have your own custom backdrop for your Scratch project!

Adding Sound:

1. Go to the **Sound** category in the Block Palette.
2. Drag the "play sound Meow until done" block to your script. You can choose different sounds or even record your own!

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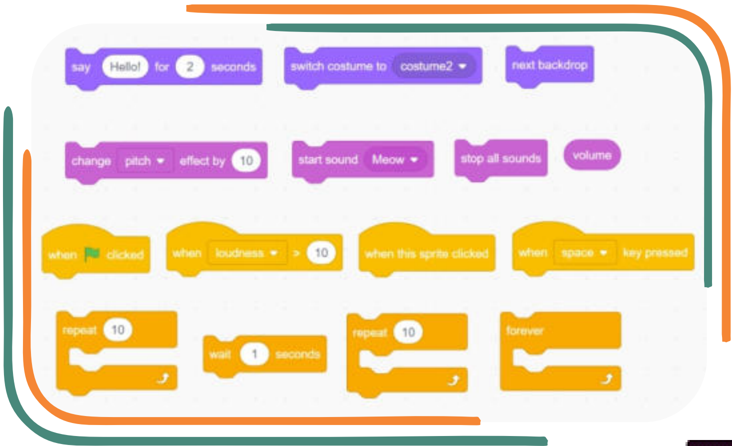
Scratch Blocks:

Here are some basic blocks you'll use often in Scratch:

- **Motion Blocks**: These blocks control how sprites move (e.g., "move 10 steps," "turn 15 degrees").
- **Looks Blocks**: These change the appearance of sprites (e.g., "say [message] for 2 seconds," "change costume").
- **Sound Blocks**: These add sound to your project (e.g., "play sound," "stop all sounds").
- **Control Blocks**: These manage the timing and looping of actions (e.g., "wait 1 second," "repeat 10 times").
- **Events Blocks**: These start actions in your script, like "when green flag clicked" or "when sprite clicked."



Creating a Simple Animation:

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Let's create a simple animation where a sprite dances.

1. Choose a Sprite: Pick a sprite like a dancer.

2. Add Costumes: Many sprites have different costumes that show different poses.

3. Change the Backdrop:

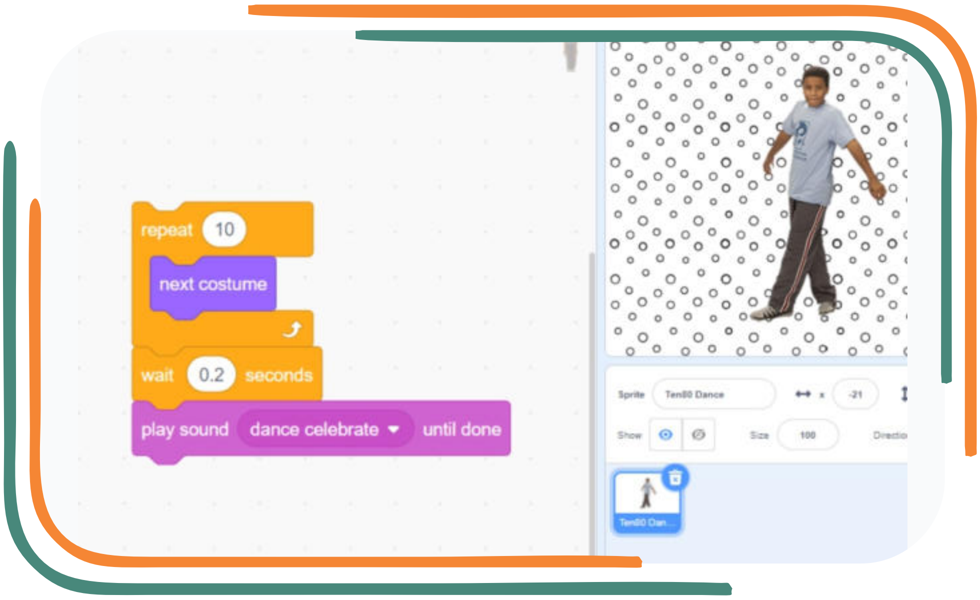
- Go to the Stage area, click on the Backdrops tab, and choose a backdrop for your animation.
- Drag the "switch backdrop to [your backdrop]" block from the Looks category to the script area.

- Place this block before the dancing actions if you want the backdrop to change at the start, or place it during the loop if you want the background to change while the sprite is dancing.

4. Create a Loop: Use a loop to switch costumes and make it look like the sprite is dancing.

- Drag the "repeat 10" block from the **Control** category to the Script Area.
- Add "next costume" from the **Looks** category inside the repeat block.
- Add "wait 0.2 seconds" to control the speed of the costume change.

5. Play Music: Add a music track from the **Sounds** category. Now, when you click the green flag, your sprite will dance to the beat!



Quick Review:

- Scratch uses blocks to represent instructions.
- Sprites are characters that you can control.
- The stage is where your sprites act out the scripts.
- Blocks are grouped by category, like Motion, Looks, and Control.

Do You Know?

Scratch was developed at MIT and is used by millions of students around the world to learn programming.

In this chapter, we learned what Scratch is and explored the main parts of its interface. We created simple scripts to make sprites move and talk and learned how to add backdrops and sounds. Now that you know the basics, you're ready to explore and create your own interactive stories and games!

EXERCISES

01: Choose the correct answer.

1. What is Scratch mainly used for?

- Reading stories
- Coding and creating projects
- Drawing pictures
- Playing games

2. In Scratch, what do you call the area where you create and run your project?

- Script Area
- Sprite Pane
- Stage
- Block Palette

3. What do blocks in Scratch represent?

- Characters
- Instructions
- Backgrounds
- None of the above

4.Which category of blocks is used to move sprites?

- Control
- Motion
- Looks
- Events

5. In Scratch, which block would you use to start a project?

- Move 10 steps
- When green flag clicked
- Say Hello!
- Forever loop

02: Fill in the blanks.

1. In Scratch, _______________ are the characters or objects that move on the stage.
2. The _________________ is where you arrange blocks to create a script.
3. The ________________ block is often used to start a project.
4. The _________________is the area where sprites perform actions.
5. In Scratch, the _______________ blocks allow you to repeat actions.

03: Write True or False for each statement.

	TRUE	FALSE
The backdrop is the character you control in Scratch.		
A sprite can be programmed to move, talk, and play sounds.		
You have to type code to make projects in Scratch.		
The script area is where you place blocks to control the sprites.		
Sprites in Scratch cannot change appearance or costumes.		

04: Answer the following questions.

1. What is the purpose of the Scratch stage?
2. How do you make a sprite move in Scratch?
3. What is a backdrop, and how is it used in Scratch?
4. How can you make a sprite say something in Scratch?
5. What is the purpose of the green flag in Scratch?

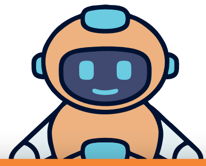
05: Match the terms with their correct definitions.

Term
1. Sprite
2. Stage
3. Script Area
4. Green Flag
5. Backdrop

Definition
a. Start a project
b. Store instructions
c. Place to display sprites
d. Characters in Scratch
e. Scene background

LAB ACTIVITIES

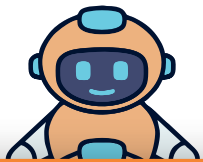
Lab Activity 1: Create a Simple Scratch Animation



Create a script where your sprite moves across the stage and says "Hello!"

1. Open **Scratch** and start a new project.
2. Select a **sprite** from the library or create your own.
3. Use the **motion** block "move 10 steps" to make the sprite move.
4. Add a suitable block to make the sprite say "Hello!"
5. Test your program by clicking the green flag.

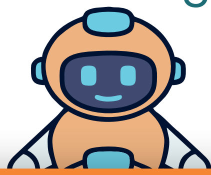
Lab Activity 2: Make a Talking Sprite



Create a script where a sprite talks and moves when the green flag is clicked.

- Open Scratch and choose any sprite.
- Go to the **Events** category and drag the "when green flag clicked" block to the Script Area.
- Go to **Motion** and add a "move 10 steps" block.
- Go to **Looks** and add a "say Hello!" block.
- Arrange the blocks in this order: "when green flag clicked," "move 10 steps," and "say Hello!"
- Click the green flag to see your sprite move and say "Hello!"

Lab Activity 3: Create a Talking Sprite Animation



Learn how to make two sprites interact with each other by using speech bubbles or sound. Use basic blocks like "say", "wait", and "broadcast".

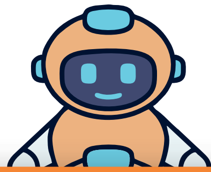
- Choose two sprites (e.g., a person and a pet).
- Make the sprites "talk" to each other using the "say" block.
- Add different speech for each sprite.
- Use "wait" blocks to control the timing between their conversations.
- Add a backdrop to create a setting for the conversation.

Lab Activity 4: Build a Simple Game with Obstacles

Learn how to create a basic game where a sprite navigates obstacles. Use "if" blocks, "motion", and "sensing" blocks.

- Choose a sprite that will be the player character.
- Create obstacles that move or appear on the screen.
- Use the "if touching [obstacle]" block to make the sprite react (e.g., losing points or restarting).
- Add sound effects when the sprite hits an obstacle.
- Make the sprite jump or move using the arrow keys or space bar.

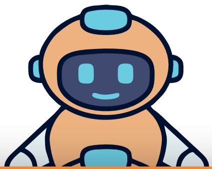
Lab Activity 5: Design a Quiz Game



Learn how to create a quiz where the user answers questions by clicking on options.

- Create a sprite that asks a question (e.g., “What is 5 + 3?”).
- Add multiple choice options as different clickable sprites.
- When an option is clicked, use "if" blocks to check if the answer is correct or not.
- Show feedback to the user using the "say" block (e.g., "Correct!" or "Try Again").
- Keep track of the score using a variable.

Lab Activity 6: Animate a Character Walking



Learn how to animate a sprite to make it appear as though it's walking or running.

- Choose a sprite and upload or create different costumes showing various walking poses.
- Use a "repeat" block to switch between costumes and create a walking animation.
- Use the "wait" block to control the speed of the animation.
- Make the sprite move across the screen using the "glide" or "move" blocks.
- Add a backdrop to create a scene for the sprite to walk through.